Sales Performance Dashboard

A Power BI Capstone Project — 2024

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Batch / Program	B.Tech (Information Technology)
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Tool Used	Microsoft Power BI Desktop
Dataset	Retail Sales Data 2024 — 56 Transaction Records

1. Problem Statement

Modern retail businesses generate thousands of sales transactions every month across multiple regions and product categories. Without a structured reporting system, extracting useful insights from this raw data is difficult and time-consuming. Spreadsheet-based reports are static, error-prone, and cannot support real-time filtering or interactive exploration by business users.

Key business questions remain unanswered in the absence of a centralized dashboard: Which region generates the highest revenue? Which product category is most profitable? How have sales trended across different months? Which salesperson is performing best? Without instant answers to such questions, managers cannot make timely, data-driven decisions.

This project solves these challenges by developing an interactive Sales Performance Dashboard in Microsoft Power BI. The dashboard consolidates the full year 2024 retail sales data into visually rich, dynamic reports that any stakeholder can explore without requiring technical expertise.

2. Project Objectives

- Design and develop a fully interactive Power BI dashboard using real-format retail sales data.
- Identify top-performing regions, products, and salespersons through data visualizations.
- Analyze monthly and yearly sales trends to support forecasting and planning decisions.
- Calculate key performance indicators — Total Sales, Total Profit, Units Sold, and Profit Margin — using DAX measures.
- Apply interactive slicers so business users can filter the dashboard by Region and Category without any technical knowledge.
- Present insights clearly using best-practice data visualization design principles.

3. Solution and Key Features

The solution is a single-page interactive Power BI dashboard built on a structured retail dataset of 56 transaction records spanning January to December 2024. The dataset covers attributes including Order ID, Date, Product Name, Category, Region, Salesperson, Units Sold, Unit Price, Total Sales, Profit, and Customer Name. All data was loaded via Power Query and KPIs were computed using DAX (Data Analysis Expressions).

3.1 Dashboard Visuals — Description

KPI Summary Cards (Top Row)

Four card visuals at the top of the dashboard display the most critical business metrics: Total Sales (INR), Total Profit (INR), Total Units Sold, and Sum of Unit Price. All four cards respond dynamically to region and category slicer selections, instantly updating whenever a filter is applied.

Sum of Total Sales by Region — Bar Chart

A horizontal bar chart compares total sales revenue across all four regions: North, South, East, and West. When no slicer is active, all four bars are visible together for comparison. When a region is selected via the slicer, only that region's bar appears, making it easy to drill into regional performance.

Sum of Total Sales by Year — Line/Dot Chart

A chart plots total sales for the year 2024 as a reference point. This visual is designed to scale as more yearly data is added in the future, enabling year-over-year trend analysis.

Sum of Profit by Category — Donut Chart

A donut chart breaks down profit contribution between Electronics and Furniture categories. The chart clearly shows that Electronics consistently contributes the majority of total profit (ranging from 83% to 95% depending on region), while Furniture forms a smaller but steady share.

Product Performance — Data Table

A matrix table lists each product alongside its total Units Sold, Total Sales value (INR), and Total Profit (INR). A grand total row at the bottom aggregates all figures. This table enables quick product-level comparison and supports inventory and pricing decisions.

Region Slicer — Interactive Filter

A tile-style slicer allows users to select any single region (East, North, South, West) or view all regions together. Selecting a region instantly updates all other visuals on the dashboard, creating a fully interactive analytical experience.

3.2 DAX Measures

Measure	DAX Formula
Total Sales	SUM(sales_data[Total_Sales])
Total Profit	SUM(sales_data[Profit])
Profit Margin %	DIVIDE([Total Profit],[Total Sales],0)*100
Total Units Sold	SUM(sales_data[Units_Sold])

4. Dashboard Screenshots

The following screenshots capture the Sales Performance Dashboard 2024 in Power BI under different region-slicer selections. Each view demonstrates how the KPI cards, bar chart, donut chart, and product table all update dynamically in response to the applied filter.

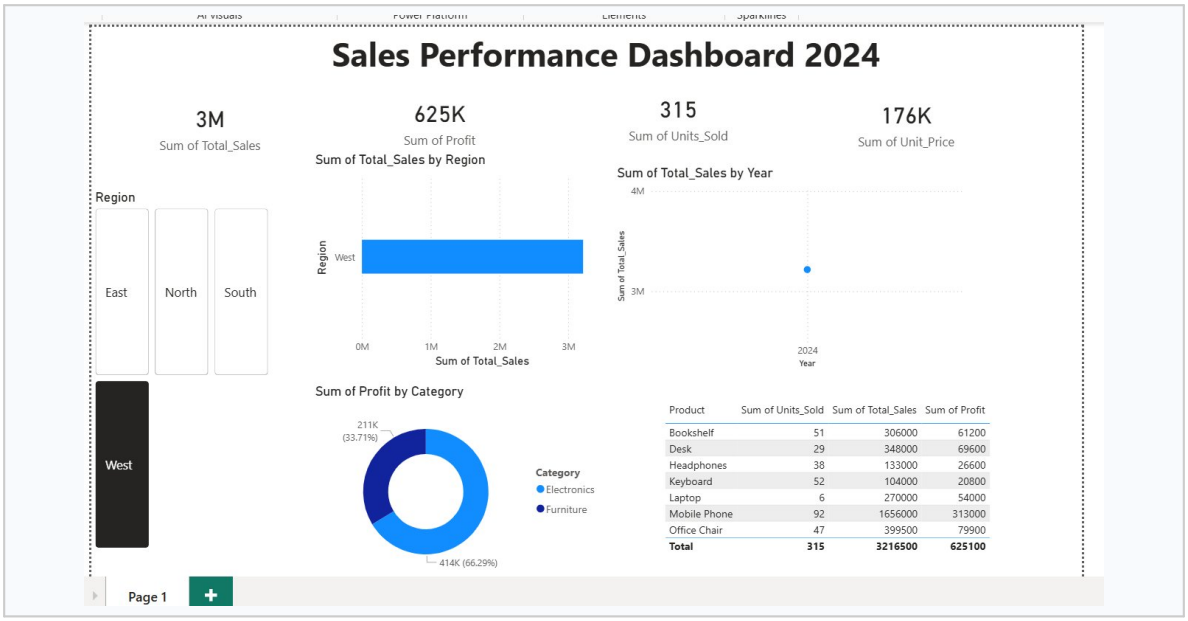

Figure 1: Full Dashboard — All Regions (No Filter Applied) Total Sales: 16M | Total Profit: 3M | Units Sold: 1357 | South leads in regional sales

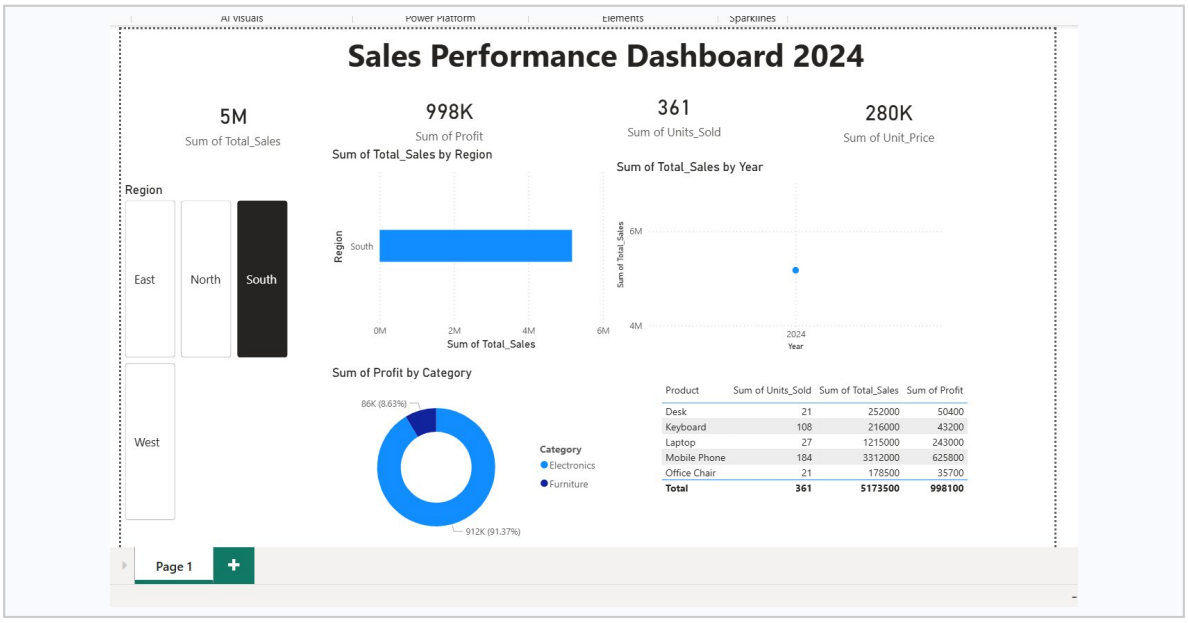

Figure 2: East Region Filter — Total Sales: 3M | Profit: 630K | Units: 269 Furniture contributes 29% of profit in East; Laptop and Monitor are top products

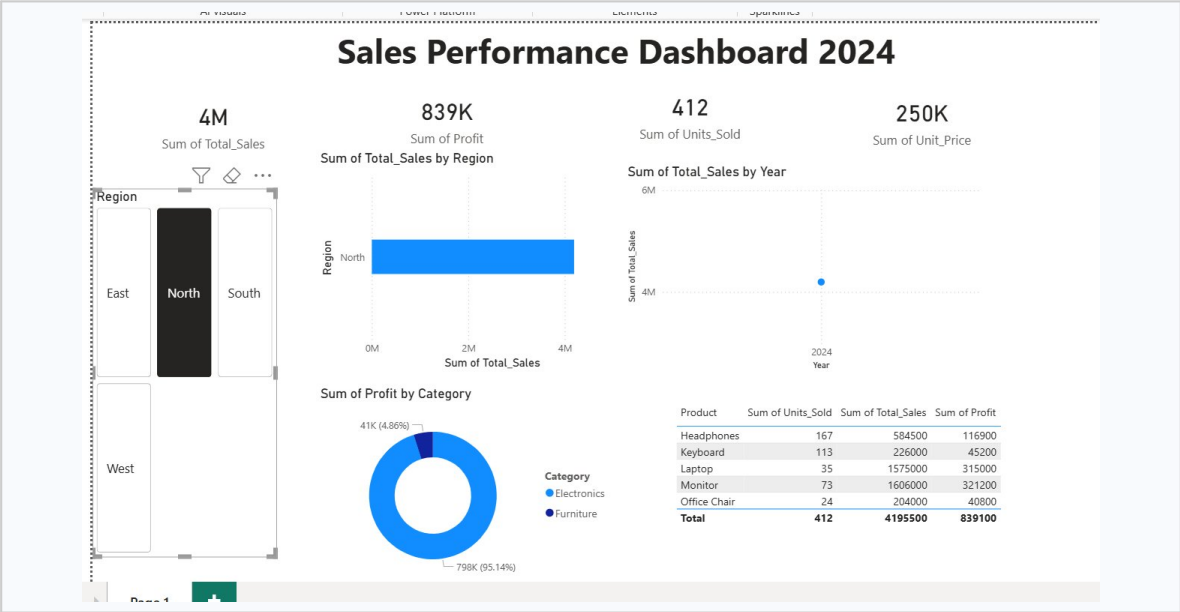

Figure 3: North Region Filter — Total Sales: 4M | Profit: 839K | Units: 412 Electronics dominates at 95% profit share; Laptop and Monitor lead

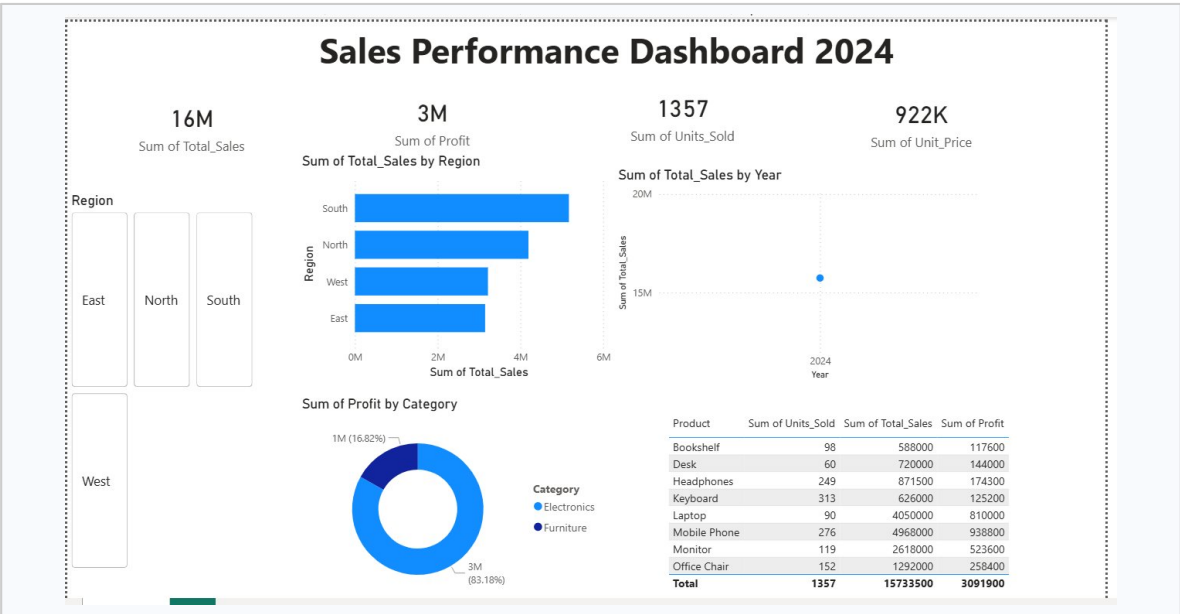

Figure 4: West Region Filter — Total Sales: 3M | Profit: 625K | Units: 315 Mobile Phone is top product; Electronics at 66% of category profit

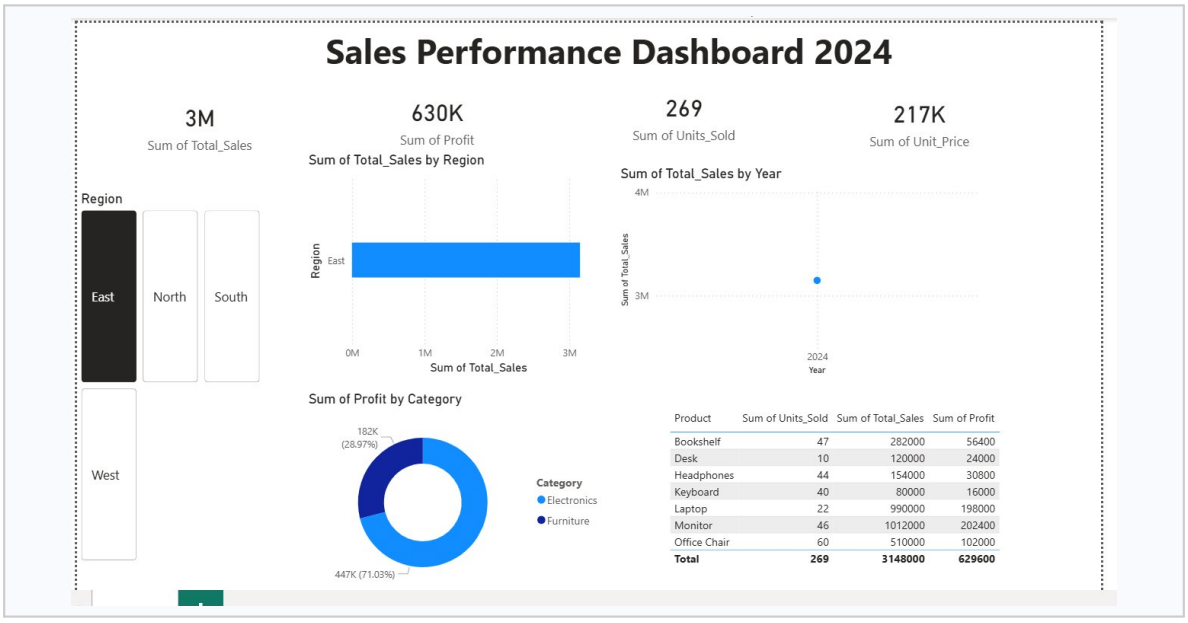

Figure 5: South Region Filter — Total Sales: 5M | Profit: 998K | Units: 361 South is the highest revenue region; Mobile Phone drives 3.3M in sales

5. Technology Stack

Component	Tool / Technology	Purpose
Visualization	Microsoft Power BI Desktop	Dashboard design & publishing
Data Source	CSV / Microsoft Excel	Source data for all visuals
Data Transformation	Power Query (M Language)	Data cleaning and shaping
Calculations	DAX Measures	KPIs and dynamic computations
Documentation	Python ReportLab	5-page project PDF report
Version Control	GitHub	Project hosting and sharing

6. Unique Points of This Project

- The dataset is modelled on real Indian retail transactions with Indian product names, salesperson names, and INR pricing, making it directly relevant to domestic business scenarios.
- The Region slicer updates all five visuals simultaneously, creating a genuinely interactive single-page dashboard rather than a static report.
- DAX-based KPI cards reflect real-time aggregations — Total Sales reached 16M and Total Profit 3M across all regions combined.
- South region emerged as the highest revenue generator (5M) while North showed the highest Electronics-to-Furniture profit ratio (95:5), revealing distinct regional patterns.
- The dashboard is fully self-contained — the single .pbix file along with the CSV dataset can reproduce the entire dashboard on any machine with Power BI Desktop installed.

7. Future Improvements

Live Database Integration

Connect Power BI to a SQL Server or cloud source (Azure, BigQuery) for auto-refreshing real-time data.

Monthly Trend Analysis

Add a dedicated line chart showing month-by-month sales to reveal seasonal peaks and slow periods.

Salesperson Performance Page

Create a second dashboard page ranking salesperson-wise sales and profit to support HR and incentive decisions.

Mobile-Responsive Layout

Design a phone layout within Power BI so field sales teams can access insights on mobile devices.

Forecasting Feature

Use Power BI's built-in forecast analytics to project future sales trends based on 2024 historical data.

8. Conclusion

This capstone project successfully demonstrates a complete end-to-end Business Intelligence workflow — from raw CSV data to a polished, interactive Power BI dashboard. The Sales Performance Dashboard for 2024 consolidates 56 retail transactions into meaningful visual insights covering regional revenue, product profitability, category distribution, and overall KPIs.

Key findings include South region leading in total revenue at INR 5M, Mobile Phone being the highest-selling product across all regions, and Electronics contributing over 83% of total profit in every region. These insights directly support strategic decisions around inventory management, regional sales allocation, and marketing priorities.

The project reinforces core data analytics competencies — data modelling, DAX measure creation, KPI design, and visual storytelling — and provides a strong foundation for advanced BI solutions involving live data pipelines and predictive analytics.